Lecture 18

Kink fluctuations

The kink is a stationary field with finite energy. We compute its energy and the quadratic change produced by a perturbation. The resulting operator factors into an adjoint pair, which determines its sign and its zero mode. The zero mode records the freedom to translate the kink.

Objectives. *We derive the kink's charge and energy, identify the domain of its fluctuation operator, and prove nonnegativity and the exact kernel. We distinguish this linear result from nonlinear orbital stability.*

By the end of this lecture you will be able to:

- Compute the kink energy and normalise its translation mode.
- Prove the adjoint factorisation on the stated Sobolev domain.
- Explain why a nonnegative Hessian alone does not prove orbital stability.

18.1 The kink and its energy

Recall the potential and stationary kink from Lecture 15:

$$V(u) = \frac{1}{2}(1-u^2)^2, \qquad b(x) = \tanh x, \qquad -b'' + V'(b) = 0.$$

We derive the energy and charge here, without assuming the corresponding calculation from *Topology and Geometry*, Part II.

Definition 18.1 (Static energy and kink charge). For a real C^1 profile ϕ with finite displayed energy integral, its *static energy* is

$$E_{\mathrm{stat}}[\phi] = \int_{\mathbb{R}} \left(\frac{1}{2}\phi_x^2 + V(\phi) \right) \mathrm{d}x.$$

If the endpoint limits exist and are vacua, its *topological charge* is $Q = [\phi(+\infty) - \phi(-\infty)]/2$.

Since $b' = \operatorname{sech}^2 x$ and $1 - b^2 = \operatorname{sech}^2 x$, the kink's two energy terms are equal. Substitution $z = \tanh x$ gives

$$Q[b] = 1, \qquad E_{\mathrm{stat}}[b] = \int_{\mathbb{R}} \operatorname{sech}^4 x \, \mathrm{d}x = \int_{-1}^{1} (1 - z^2) \, \mathrm{d}z = \frac{4}{3}. \tag{18.1}$$

This is a minimum among finite-energy C^1 profiles with limits -1 and 1. Indeed, write $W(u) = u - u^3/3$, so $V = \frac{1}{2}(W')^2$. Completing the square on finite intervals and taking the endpoint limits yields

$$E_{\mathrm{stat}}[\phi] = \frac{1}{2} \int_{\mathbb{R}} [\phi_x - (1 - \phi^2)]^2 \, \mathrm{d}x + W(1) - W(-1) \geq \frac{4}{3}.$$

The cross term is integrable by Cauchy–Schwarz and finite energy. The kink saturates the bound because $b' = 1 - b^2$.

18.2 The fluctuation operator

For a real compact smooth perturbation h, expand the energy of $b+\varepsilon h$. The first variation has boundary term $[b'h]_{-R}^{R} = 0$ on an interval containing the compact support of h. Integration by parts and the stationary equation make its remaining integral zero. The second variation is

$$\frac{\mathrm{d}^2}{\mathrm{d}\varepsilon^2} E_{\mathrm{stat}}[b+\varepsilon h]\Big|_{\varepsilon=0} = \int_{\mathbb{R}} \left[h_x^2 + V''(b)h^2\right] \mathrm{d}x, \qquad V''(b) = 4 - 6\,\mathrm{sech}^2 x. \tag{18.2}$$

The operator suggested by integration by parts is

$$H_K = -\partial_x^2 + 4 - 6\,\mathrm{sech}^2 x. \tag{18.3}$$

The coefficient is negative near the origin, as shown in Figure 18.1. Its pointwise sign does not determine the sign of the full quadratic expression, which also contains h_x^2.

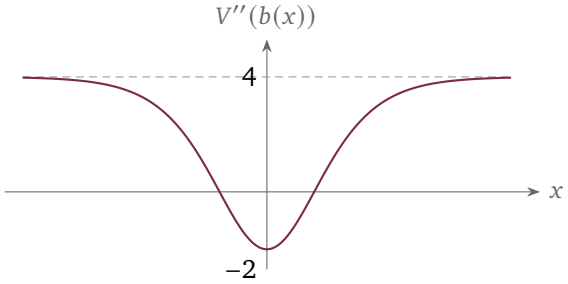


Figure 18.1: The fluctuation potential $4 - 6\,\mathrm{sech}^2 x$ has value -2 at the centre and tends to 4 at both spatial ends.

Linearising the field equation at b gives $h_{tt} + H_K h = 0$. The same operator therefore describes the energy Hessian and the linearised motion. We now assign it an operator domain.

For adjoints, use the complex Hilbert space $L^2(\mathbb{R})$ with inner product $(u, v) = \int \overline{u}v$, antilinear in the first slot. Physical perturbations are real. In addition to Lecture 4's H^1, define

$$H^2(\mathbb{R}) = \{u \in L^2 : (1+k^2)\widehat{u} \in L^2\}, \qquad \|u\|_{H^2} = \|(1+k^2)\widehat{u}\|_2.$$

By the Fourier derivative rule, this is exactly the class with first and second weak derivatives in L^2.

We need approximation and integration by parts on these spaces. For $s = 0, 1, 2$, with $H^0 = L^2$, compact smooth functions are dense in H^s. To see this, multiply u by a smooth cutoff $\chi(x/R)$. The product rule expresses the error in each derivative as a tail of an L^2 derivative of u plus terms bounded by C/R or C/R^2 times lower derivatives. Each tends to zero in L^2. Then convolve the compactly supported result with a compact smooth unit-mass mollifier. Convolution commutes with weak derivatives, and translation continuity in L^2 gives convergence of each derivative. Translation continuity follows from Plancherel and dominated convergence applied to $e^{ikh}\widehat{u}(k)$ as $h \to 0$. This proves the claimed density and, by passage from compact tests,

$$(u', v) = -(u, v') \qquad (u, v \in H^1). \tag{18.4}$$

The quadratic form from (18.2) consequently extends to H^1. It is also the energy Hessian there: the remaining terms in the exact quartic expansion are cubic and quartic in h. They are integrable and continuous in H^1 by Lecture 15's $H^1 \to L^\infty$ bound.

Definition 18.2 (First-order operator and adjoint)**.** Let $A : H^1 \subset L^2 \to L^2$ be given by

$$Au = u' + 2\tanh x\, u. \tag{18.5}$$

The domain of its Hilbert adjoint $A^\dagger$ consists of the $v \in L^2$ for which there exists $g \in L^2$ satisfying $(Au, v) = (u, g)$ for every $u \in H^1$. Its value is $A^\dagger v = g$.

Density of H^1 in L^2 makes g unique. A *formal differential identity* only compares expressions on smooth tests. An identity of unbounded operators must also identify their domains.

Proposition 18.3 (Kink factorisation and kernel)**.** *The Hilbert adjoint has domain H^1 and expression $A^\dagger = -\partial_x + 2\tanh x$. The operator product has domain H^2, and*

$$H_K = A^\dagger A : H^2(\mathbb{R}) \subset L^2(\mathbb{R}) \longrightarrow L^2(\mathbb{R}), \qquad (u, H_K u) = \|Au\|_2^2 \geq 0. \tag{18.6}$$

Its kernel is

$$\ker H_K = \operatorname{span}\{\operatorname{sech}^2 x\}. \tag{18.7}$$

The quadratic-form factorisation also holds on H^1:

$$\mathfrak{q}[u] = \int_{\mathbb{R}} \left[|u'|^2 + (4 - 6\operatorname{sech}^2 x)|u|^2\right] \mathrm{d}x = \|u' + 2\tanh x\, u\|_2^2. \tag{18.8}$$

Proof. First, integration by parts (18.4) shows that every $v \in H^1$ is in $D(A^\dagger)$, with the stated expression. Conversely, if $v \in D(A^\dagger)$ with value g, testing against $u \in C_c^\infty$ gives $-v' + 2\tanh x\, v = g$ distributionally. The bounded coefficient implies $v' \in L^2$, hence $v \in H^1$. This identifies the actual adjoint domain.

Multiplication by $w = 2\tanh x$ preserves H^1, since w and w' are bounded and $(wu)' = w'u + wu'$ by approximation. Thus $u \in H^2$ implies $Au \in H^1$. Conversely, if $u \in H^1$ and $Au \in H^1$, then $u' = Au - wu \in H^1$, so $u \in H^2$. This proves $D(A^\dagger A) = H^2$.

On a compact smooth test, the product rule gives

$$(-\partial_x + w)(\partial_x + w)u = -u'' + (w^2 - w')u = -u'' + (4 - 6\operatorname{sech}^2 x)u.$$

All terms depend continuously on u in H^2, so density proves the operator identity on that domain. The adjoint relation now gives $(u, H_K u) = (Au, Au)$. Likewise, expansion and compact integration by parts give $\|u' + wu\|_2^2 = \|u'\|_2^2 + \int (w^2 - w')|u|^2$. Density in H^1 proves (18.8).

Finally, $H_K u = 0$ forces $Au = 0$. An H^1 function has a locally absolutely continuous representative: subtracting the primitive of its weak derivative leaves a distribution with derivative zero, hence a constant on each interval. The integrating factor therefore gives

$$(\cosh^2 x\, u(x))' = \cosh^2 x\, Au(x) = 0.$$

It follows that $u = C\operatorname{sech}^2 x$. This profile and its first two derivatives decay exponentially, so it belongs to H^2 and directly satisfies $Au = H_K u = 0$. Thus it gives the whole kernel. ∎

The nonnegativity assertion means the quadratic inequality in (18.6) on the stated domain. The proof uses the adjoint and the form identity, without requiring a spectral theorem.

18.3 Normalising the translation mode

Equation (18.1) already gives the squared norm of the kernel profile. Hence its positive unit vector is

$$\psi_0(x) = \frac{\sqrt{3}}{2}\operatorname{sech}^2 x, \qquad \|\psi_0\|_2 = 1. \tag{18.9}$$

For the translated family $b_{x_0}(x) = \tanh(x - x_0)$,

$$\partial_{x_0} b_{x_0}(x)\big|_{x_0=0} = -\operatorname{sech}^2 x = -\frac{2}{\sqrt{3}}\psi_0(x).$$

Thus translation generates exactly the one-dimensional kernel. Differentiating the stationary equation along this family also gives $H_K b' = 0$, in agreement with the factorisation.

There are nonzero positive directions as well. For $u(x) = \operatorname{sech} x$, $Au = \operatorname{sech} x \tanh x$. The same substitution $z = \tanh x$ gives

$$(u, H_K u) = \|Au\|_2^2 = \int_{\mathbb{R}} \operatorname{sech}^2 x \tanh^2 x \, \mathrm{d}x = \int_{-1}^{1} z^2 \, \mathrm{d}z = \frac{2}{3}.$$

The negative coefficient near $x = 0$ has not produced a negative form value.

18.4 What the linear statement establishes

No exponentially growing normal mode of the form $h(t, x) = e^{\gamma t}u(x)$, with $\gamma > 0$ and nonzero $u \in H^2$, can solve the linearised equation. Substitution and the factorisation give

$$0 = (u, (\gamma^2 + H_K)u) = \gamma^2\|u\|_2^2 + \|Au\|_2^2,$$

which is impossible for such u. This is a statement about the linear equation. It is not a bound on every linearised displacement: the exact solution $h(t) = t\psi_0$ has growing H^1 norm, although its quadratic energy $\frac{1}{2}(\|h_t\|_2^2 + \mathfrak{q}[h]) = 1/2$ is constant. It describes drift in the translation direction to first order.

Each $b_{x_0} - b$ belongs to H^1: write the difference as an integral of translated b', and use the same argument with b'' for its derivative. On the affine class $\phi - b \in H^1$ and $\pi = \phi_t \in L^2$, define distance to the static kink family by

$$d_{\mathrm{orb}}(\phi, \pi) = \inf_{x_0 \in \mathbb{R}} \big(\|\phi - b_{x_0}\|_{H^1} + \|\pi\|_2\big).$$

Orbital stability would require that for every $\varepsilon > 0$ there is $\delta > 0$ such that data with this distance below δ have distance below ε for all future times. The factorisation does not prove this nonlinear assertion. It does not estimate the higher-order energy terms or choose a translating centre during nonlinear evolution. Theorem 16.3 supplies global energy-class evolution, but its continuation estimates allow growth and do not give this distance bound. Lecture 27 returns to exact coherent solutions and their parameters.

Summary. The kink has charge +1 and static energy 4/3. Its energy Hessian and linearised equation use $H_K = -\partial_x^2 + 4 - 6\operatorname{sech}^2 x$. The actual Hilbert adjoint of $A = \partial_x + 2\tanh x$ has domain H^1, and $A^\dagger A = H_K$ has domain H^2. The factorisation gives a nonnegative quadratic form and the exact translation kernel, with unit mode $\sqrt{3}\operatorname{sech}^2 x/2$. These linear conclusions do not establish nonlinear orbital stability.

Self-check. Reverse the factor order. Is $AA^\dagger$ equal to H_K?

Answer. No. The derivative of $2\tanh x$ changes sign in the product, giving $AA^\dagger = -\partial_x^2 + 4 - 2\operatorname{sech}^2 x$. The order is essential.

Self-check. Why is H^1 the form domain used here, while H^2 is the operator domain?

Answer. The form needs one derivative in L^2. Applying the differential operator requires a second derivative in L^2. The product domain proof shows that $u \in H^1$ and $Au \in H^1$ are exactly $u \in H^2$.

Self-check. Normalise $b'(x)$ in L^2 and compare it with the derivative of $b_{x_0}(x)$ with respect to x_0 at zero.

Answer. Since $\|b'\|_2^2 = 4/3$, its positive unit normalisation is $\psi_0 = (\sqrt{3}/2)b'$. The translation derivative is $-b' = -(2/\sqrt{3})\psi_0$.

Self-check. Does the exact linear solution $h = t\psi_0$ contradict nonnegativity? Does it by itself disprove nonlinear orbital stability?

Answer. Neither. Its quadratic energy is the constant $1/2$, while the form gives no restoring term in the translation direction. The solution belongs to the linearised equation. Nonlinear orbital stability concerns distance to the entire translated kink family.